

CHAPTER C11

SEISMIC DESIGN CRITERIA

C11.1 GENERAL

Many of the technical changes made to the seismic provisions of the 2010 edition of this standard are primarily based on Part 1 of the 2009 edition of the *NEHRP Recommended Seismic Provisions for New Buildings and Other Structures* (FEMA 2009), which was prepared by the Building Seismic Safety Council (BSSC) under sponsorship of the Federal Emergency Management Agency (FEMA) as part of its contribution to the National Earthquake Hazards Reduction Program (NEHRP). The National Institute of Standards and Technology (NIST) is the lead agency for NEHRP, the federal government's long-term program to reduce the risks to life and property posed by earthquakes in the United States. Since 1985, the NEHRP provisions have been updated every three to five years. The efforts by BSSC to produce the NEHRP provisions were preceded by work performed by the Applied Technology Council (ATC) under sponsorship of the National Bureau of Standards (NBS)—now NIST—which originated after the 1971 San Fernando Valley earthquake. These early efforts demonstrated the design rules of that time for seismic resistance but had some serious shortcomings. Each subsequent major earthquake has taught new lessons. The NEHRP agencies (FEMA, NIST, the National Science Foundation [NSF], and the U.S. Geological Survey [USGS]), ATC, BSSC, ASCE, and others have endeavored to work individually and collectively to improve each succeeding document to provide state-of-the-art earthquake engineering design and construction provisions and to ensure that the provisions have nationwide applicability.

Content of Commentary. This commentary is updated from the enhanced commentary to ASCE/SEI 7-10 that was based substantially on Part 2, Commentary, of the 2009 *NEHRP Recommended Seismic Provisions for New Buildings and Other Structures* (FEMA 2009). For additional background on the earthquake provisions contained in Chapters 11 through 23 of ASCE/SEI 7-10, the reader is referred to *Recommended Lateral Force Requirements and Commentary* (SEAOC 1999).

Nature of Earthquake “Loads.” Earthquakes load structures indirectly through ground motion. As the ground shakes, a structure responds. The response vibration produces structural deformations with associated strains and stresses. The computation of dynamic response to earthquake ground shaking is complex. The design forces prescribed in this standard are intended only as approximations to generate internal forces suitable for proportioning the strength and stiffness of structural elements and for estimating the deformations (when multiplied by the deflection amplification factor, C_d) that would occur in the same structure in the event of the design-level earthquake ground motion (not MCE_R).

The basic methods of analysis in the standard use the common simplification of a response spectrum. A response spectrum for a

specific earthquake ground motion provides the maximum value of response for elastic single-degree-of-freedom oscillators as a function of period without the need to reflect the total response history for every period of interest. The design response spectrum specified in Section 11.4 and used in the basic methods of analysis in Chapter 12 is a smoothed and normalized approximation for many different recorded ground motions.

The design limit state for resistance to an earthquake is unlike that for any other load within the scope of ASCE 7. The earthquake limit state is based upon system performance, not member performance, and considerable energy dissipation through repeated cycles of inelastic straining is assumed. The reason is the large demand exerted by the earthquake and the associated high cost of providing enough strength to maintain linear elastic response in ordinary buildings. This unusual limit state means that several conveniences of elastic behavior, such as the principle of superposition, are not applicable and makes it difficult to separate design provisions for loads from those for resistance. This difficulty is the reason Chapter 14 of the standard contains so many provisions that modify customary requirements for proportioning and detailing structural members and systems. It is also the reason for the construction quality assurance requirements.

Use of Allowable Stress Design Standards. The conventional design of almost all masonry structures and many wood and steel structures has been accomplished using allowable stress design (ASD). Although the fundamental basis for the earthquake loads in Chapters 11 through 23 is a strength limit state beyond the first yield of the structure, the provisions are written such that conventional ASD methods can be used by the design engineer. Conventional ASD methods may be used in one of two ways:

1. The earthquake load as defined in Chapters 11 through 23 may be used directly in allowable stress load combinations of Section 2.4, and the resulting stresses may be compared directly with conventional allowable stresses.
2. The earthquake load may be used in strength design load combinations, and resulting stresses may be compared with amplified allowable stresses (for those materials for which the design standard gives the amplified allowable stresses, e.g., masonry).

Federal Government Construction. The Interagency Committee on Seismic Safety in Construction has prepared an order executed by the president (Executive Order 12699 2016) that all federally owned or leased building construction, as well as federally regulated and assisted construction, should be constructed to mitigate seismic hazards and that the NEHRP provisions are deemed to be the suitable standard. It is expected that this standard would be deemed equivalent, but the reader should bear in mind that there are certain differences.

C11.1.1 Purpose. The purpose of Section 11.1.1 is to clarify that the detailing requirements and limitations prescribed in this section and referenced standards are still required even when the design load combinations involving the wind forces of Chapters 26 through 29 produce greater effects than the design load combinations involving the earthquake forces of Chapters 11 through 23. This detailing is required so that the structure resists, in a ductile manner, potential seismic loads in excess of the prescribed wind loads. A proper, continuous load path is an obvious design requirement, but experience has shown that it is often overlooked and that significant damage and collapse can result. The basis for this design requirement is twofold:

1. To ensure that the design has fully identified the seismic force-resisting system and its appropriate design level and
2. To ensure that the design basis is fully identified for the purpose of future modifications or changes in the structure.

Detailed requirements for analyzing and designing this load path are given in the appropriate design and materials chapters.

C11.1.2 Scope. Certain structures are exempt for the following reasons:

Exemption 1—Detached wood-frame dwellings not exceeding two stories above grade plane constructed in accordance with the prescriptive provisions of the International Residential Code (IRC) for light-frame wood construction, including all applicable IRC seismic provisions and limitations, are deemed capable of resisting the anticipated seismic forces. Detached one- and two-story wood-frame dwellings generally have performed well even in regions of higher seismicity. Therefore, within its scope, the IRC adequately provides the level of safety required for buildings. The structures that do not meet the prescriptive limitations of the IRC are required to be designed and constructed in accordance with the International Building Code (IBC) and the ASCE 7 provisions adopted therein.

Exemption 2—Agricultural storage structures generally are exempt from most code requirements because such structures are intended only for incidental human occupancy and represent an exceptionally low risk to human life.

Exemption 3—Bridges, transmission towers, nuclear reactors, and other structures with special configurations and uses are not covered. The regulations for buildings and buildinglike structures presented in this document do not adequately address the design and performance of such special structures.

ASCE 7 is not retroactive and usually applies to existing structures only when there is an addition, change of use, or alteration. Minimum acceptable seismic resistance of existing buildings is a policy issue normally set by the authority having jurisdiction. Appendix 11B of the standard contains rules of application for basic conditions. ASCE 41 (2014) provides technical guidance but does not contain policy recommendations. A chapter in the *International Building Code* (IBC) applies to alteration, repair, addition, and change of occupancy of existing buildings, and the International Code Council maintains the *International Existing Building Code* (IEBC) and associated commentary.

C11.1.3 Applicability. Industrial buildings may be classified as nonbuilding structures in certain situations for the purposes of determining seismic design coefficients and factors, system limitations, height limits, and associated detailing requirements. Many industrial building structures have geometries and/or framing systems that are different from the broader class of occupied structures addressed by Chapter 12, and the limited nature of the occupancy associated with these buildings reduces the hazard associated with their performance in earthquakes. Therefore, when the occupancy is limited primarily to

maintenance and monitoring operations, these structures may be designed in accordance with the provisions of Section 15.5 for nonbuilding structures similar to buildings. Examples of such structures include, but are not limited to, boiler buildings, aircraft hangars, steel mills, aluminum smelting facilities, and other automated manufacturing facilities, whereby the occupancy restrictions for such facilities should be uniquely reviewed in each case. These structures may be clad or open structures.

C11.1.4 Alternate Materials and Methods of Construction. It is not possible for a design standard to provide criteria for the use of all possible materials and their combinations and methods of construction, either existing or anticipated. This section serves to emphasize that the evaluation and approval of alternate materials and methods require a recognized and accepted approval system. The requirements for materials and methods of construction contained within the document represent the judgment of the best use of the materials and methods based on well-established expertise and historical seismic performance. It is important that any replacement or substitute be evaluated with an understanding of all the ramifications of performance, strength, and durability implied by the standard.

Until needed standards and agencies are created, authorities that have jurisdiction need to operate on the basis of the best evidence available to substantiate any application for alternates. If accepted standards are lacking, applications for alternative materials or methods should be supported by test data obtained from test data requirements in Section 1.3.1. The tests should simulate expected load and deformation conditions to which the system, component, or assembly may be subjected during the service life of the structure. These conditions, when applicable, should include several cycles of full reversals of loads and deformations in the inelastic range.

C11.1.5 Quality Assurance. Quality assurance (QA) requirements are essential for satisfactory performance of structures in earthquakes. QA requirements are usually incorporated in building codes as special inspections and tests or as structural observation, and they are enforced through the authorities having jurisdiction. Many building code requirements parallel or reference the requirements found in standards adopted by ASCE 7. Where special inspections and testing, or structural observations are not specifically required by the building code, a level of quality assurance is usually provided by inspectors employed by the authority having jurisdiction.

Where building codes are not in force or where code requirements do not apply to or are inadequate for a unique structure or system, the registered design professional for the structure or system should develop a QA program to verify that the structure or system is constructed as designed. A QA program could be modeled on similar provisions in the building code or applicable standards.

The quality assurance plan is used to describe the QA program to the owner, the authority having jurisdiction, and to all other participants in the QA program. As such, the QA plan should include definitions of roles and responsibilities of the participants. It is anticipated that in most cases the owner of the project would be responsible for implementing the QA plan.

C11.2 DEFINITIONS

ATTACHMENTS, COMPONENTS, AND SUPPORTS: The distinction among attachments, components, and supports is necessary to the understanding of the requirements for nonstructural components and nonbuilding structures. Common cases associated with nonstructural elements are illustrated in Fig. C11.2-1. The

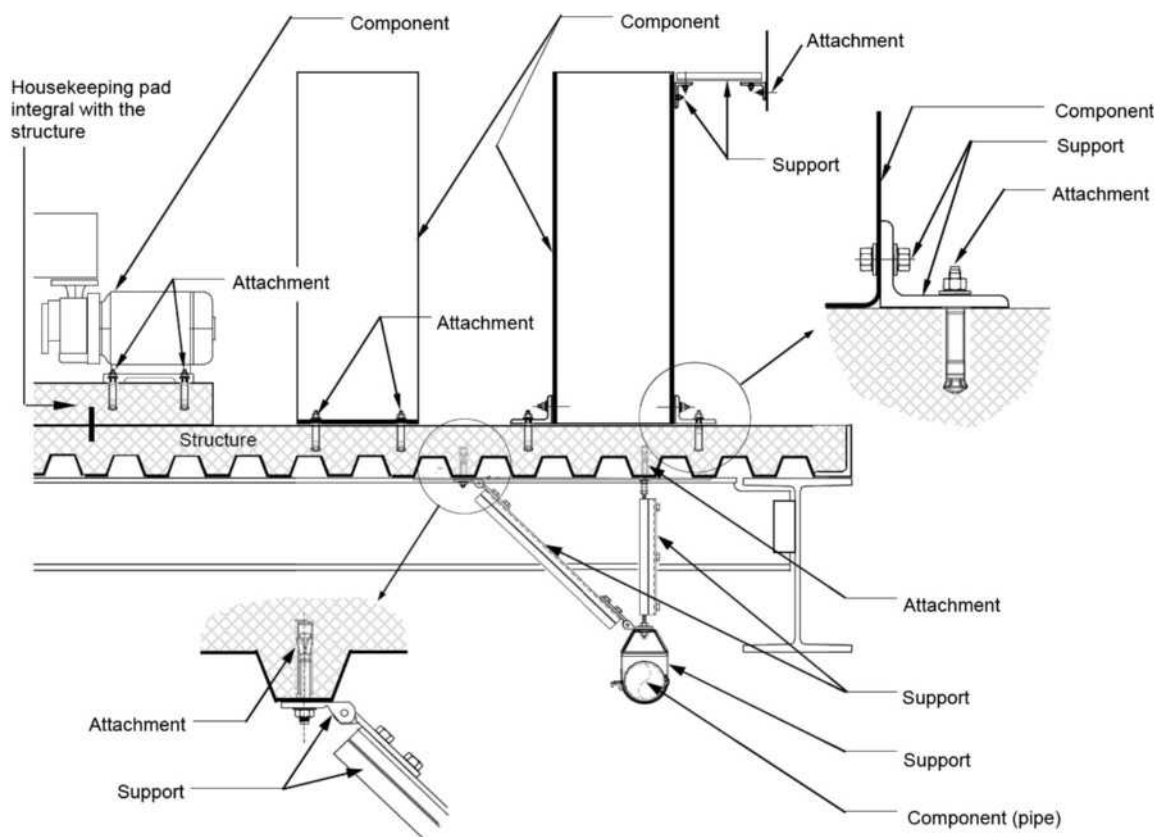


FIGURE C11.2-1 Examples of Attachments, Components, and Supports

definitions of attachments, components, and supports are generally applicable to components with a defined envelope in the as-manufactured condition and for which additional supports and attachments are required to provide support in the as-built condition. This distinction may not always be clear, particularly when the component is equipped with prefabricated supports; therefore, judgment must be used in the assignment of forces to specific elements in accordance with the provisions of Chapter 13.

BASE: The following factors affect the location of the seismic base:

- location of the grade relative to floor levels,
- soil conditions adjacent to the building,
- openings in the basement walls,
- location and stiffness of vertical elements of the seismic force-resisting system,
- location and extent of seismic separations,
- depth of basement,
- manner in which basement walls are supported,
- proximity to adjacent buildings, and
- slope of grade.

For typical buildings on level sites with competent soils, the base is generally close to the grade plane. For a building without a basement, the base is generally established near the ground-level slab elevation, as shown in Fig. C11.2-2. Where the vertical elements of the seismic force-resisting system are supported on interior footings or pile caps, the base is the top of these elements. Where the vertical elements of the seismic force-resisting system are supported on top of perimeter foundation walls, the base is typically established at the top of the foundation walls. Often vertical elements are supported at various elevations on the top of

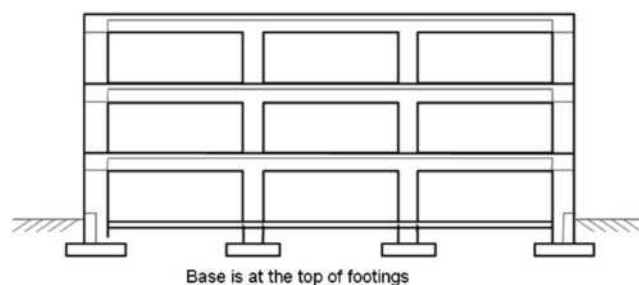


FIGURE C11.2-2 Base for a Level Site

footings, pile caps, and perimeter foundation walls. Where this occurs, the base is generally established as the lowest elevation of the tops of elements supporting the vertical elements of the seismic force-resisting system.

For a building with a basement located on a level site, it is often appropriate to locate the base at the floor closest to grade, as shown in Fig. C11.2-3. If the base is to be established at the level located closest to grade, the soil profile over the depth of the basement should not be liquefiable in the MCE_G ground motion. The soil profile over the depth of the basement also should not include quick and highly sensitive clays or weakly cemented soils prone to collapse in the MCE_G ground motion. Where liquefiable soils or soils susceptible to failure or collapse in an MCE_G ground motion are located within the depth of the basement, the base may need to be located below these soils rather than close to grade. Stiff soils are required over the depth of the basement because seismic forces are transmitted to and



FIGURE C11.2-3 Base at Ground Floor Level

from the building at this level and over the height of the basement walls. The engineer of record is responsible for establishing whether the soils are stiff enough to transmit seismic forces near grade. For tall or heavy buildings or where soft soils are present within the depth of the basement, the soils may compress laterally too much during an earthquake to transmit seismic forces near grade. For these cases, the base should be located at a level below grade.

In some cases, the base may be at a floor level above grade. For the base to be located at a floor level above grade, stiff foundation walls on all sides of the building should extend to the underside of the elevated level considered the base. Locating the base above grade is based on the principles for the two-stage equivalent lateral force procedure for a flexible upper portion of a building with one-tenth the stiffness of the lower portion of the building, as permitted in Section 12.2.3.2. For a floor level above grade to be considered the base, it generally should not be above grade more than one-half the height of the basement story, as shown in Fig. C11.2-4. Fig. C11.2-4 illustrates the concept of the base level located at the top of a floor level above grade, which also includes light-frame floor systems that rest on top of stiff basement walls or stiff crawl space stem walls of concrete or masonry construction.

A condition where the basement walls that extend above grade on a level site may not provide adequate stiffness is where the basement walls have many openings for items such as light wells, areaways, windows, and doors, as shown in Fig. C11.2-5. Where the basement wall stiffness is inadequate, the base should be taken as the level close to but below grade. If all of the vertical elements of the seismic force-resisting system are located on top of basement walls and there are many openings in the basement walls, it may be appropriate to establish the base at the bottom of the openings. Another condition where the basement walls may



FIGURE C11.2-4 Base at Level Closest to Grade Elevation

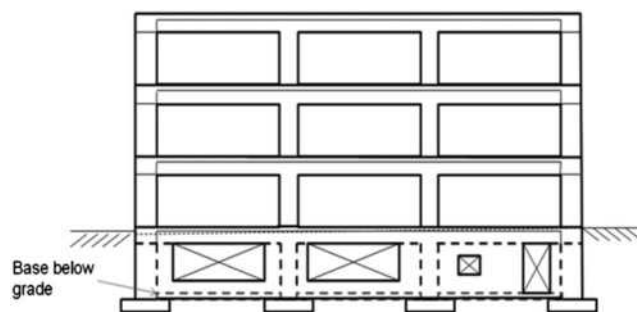


FIGURE C11.2-5 Base Below Substantial Openings in Basement Wall

not be stiff enough is where the vertical elements of the seismic force-resisting system are long concrete shear walls extending over the full height and length of the building, as shown in Fig. C11.2-6. For this case, the appropriate location for the base is the foundation level of the basement walls.

Where the base is established below grade, the weight of the portion of the story above the base that is partially above and below grade must be included as part of the effective seismic weight. If the equivalent lateral force procedure is used, this procedure can result in disproportionately high forces in upper levels because of a large mass at this lowest level above the base. The magnitude of these forces can often be mitigated by using the two-stage equivalent lateral force procedure where it is allowed or by using dynamic analysis to determine force distribution over the height of the building. If dynamic analysis is used, it may be necessary to include multiple modes to capture the required mass participation, unless soil springs are incorporated into the model. Incorporation of soil springs into the model generally reduces seismic forces in the upper levels. With one or more stiff stories below more flexible stories, the dynamic behavior of the structure may result in the portion of the base shear from the first mode being less than the portion of base shear from higher modes.

Other conditions may also necessitate establishing the base below grade for a building with a basement that is located on a level site. Such conditions include those where seismic separations extend through all floors, including those located close to and below grade; those where the floor diaphragms close to and below grade are not tied to the foundation wall; those where the floor diaphragms, including the diaphragm for the floor close to grade, are flexible; and those where other buildings are located nearby.

For a building with seismic separations extending through the height of the building including levels close to and below grade, the separate structures are not supported by the soil against a

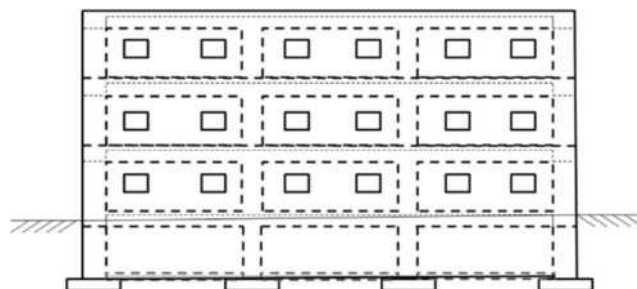


FIGURE C11.2-6 Base at Foundation Level Where There Are Full-Length Exterior Shear Walls

basement wall on all sides in all directions. If there is only one joint through the building, assigning the base to the level close to grade may still be appropriate if the soils over the depth of the basement walls are stiff and the diaphragm is rigid. Stiff soils are required so that the seismic forces can be transferred between the soils and basement walls in both bearing and side friction. If the soils are not stiff, adequate side friction may not develop for movement in the direction perpendicular to the joint.

For large footprint buildings, seismic separation joints may extend through the building in two directions and there may be multiple parallel joints in a given direction. For individual structures within these buildings, substantial differences in the location of the center of rigidity for the levels below grade relative to levels above grade can lead to torsional response. For such buildings, the base should usually be at the foundation elements below the basement or the highest basement slab level where the separations are no longer provided.

Where floor levels are not tied to foundation walls, the base may need to be located well below grade at the foundation level. An example is a building with tieback walls and posttensioned floor slabs. For such a structure, the slabs may not be tied to the wall to allow relative movement between them. In other cases, a soft joint may be provided. If shear forces cannot be transferred between the wall and a ground level or basement floor, the location of the base depends on whether forces can be transferred through bearings between the floor diaphragm and basement wall and between the basement wall and the surrounding soils. Floor diaphragms bearing against the basement walls must resist the compressive stress from earthquake forces without buckling. If a seismic or expansion joint is provided in one of these buildings, the base almost certainly needs to be located at the foundation level or a level below grade where the joint no longer exists.

If the diaphragm at grade is flexible and does not have substantial compressive strength, the base of the building may need to be located below grade. This condition is more common with existing buildings. Newer buildings with flexible diaphragms should be designed for compression to avoid the damage that can otherwise occur.

Proximity to other structures can also affect where the base should be located. If other buildings with basements are located adjacent to one or more sides of a building, it may be appropriate to locate the base at the bottom of the basement. The closer the adjacent building is to the building, the more likely it is that the base should be below grade.

For sites with sloping grade, many of the same considerations for a level site are applicable. For example, on steeply sloped sites, the earth may be retained by a tieback wall so that the building does not have to resist the lateral soil pressures. For such a case, the building is independent of the wall, so the base should be located at a level close to the elevation of grade on the side of the building where it is lowest, as shown in Fig. C11.2-7. Where

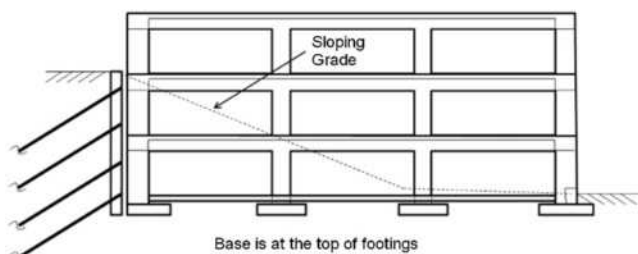


FIGURE C11.2-7 Building with Tie-Back or Cantilevered Retaining Wall That Is Separate from the Building

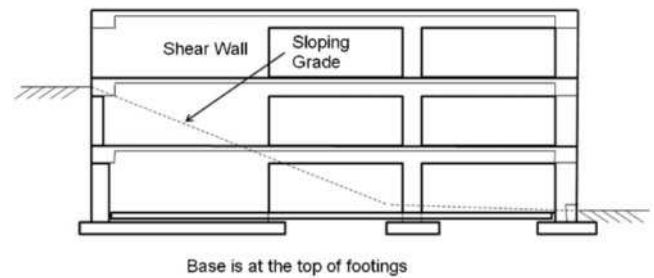


FIGURE C11.2-8 Building with Vertical Elements of the Seismic Force-Resisting System Supporting Lateral Earth Pressures

the building's vertical elements of the seismic force-resisting system also resist lateral soil pressures, as shown in Fig. C11.2-8, the base should also be located at a level close to the elevation of grade on the side of the building where grade is low. For these buildings, the seismic force-resisting system below highest grade is often much stiffer than the system used above it, as shown in Fig. C11.2-9, and the seismic weights for levels close to and below highest grade are greater than for levels above highest grade. Use of a two-stage equivalent lateral force procedure can be useful for these buildings.

Where the site is moderately sloped such that it does not vary in height by more than a story, stiff walls often extend to the underside of the level close to the elevation of high grade, and the seismic force-resisting system above grade is much more flexible above grade than it is below grade. If the stiff walls extend to the underside of the level close to high grade on all sides of the building, locating the base at the level closest to high grade may be appropriate. If the stiff lower walls do not extend to the underside of the level located closest to high grade on all sides of the building, the base should be assigned to the level closest to low grade. If there is doubt as to where to locate the base, it should conservatively be taken at the lower elevation.

DISTRIBUTION SYSTEM: For the purposes of determining the anchorage of components in Chapter 13, a distribution system is characterized as a series of individual in-line mechanical or electrical components that have been physically attached together to function as an interconnected system. In general, the individual in-line components of a distribution system are comparable to those of the pipe, duct, or electrical raceway so that the overall seismic behavior of the system is relatively uniform along its length. For example, a damper in a duct or a valve in a pipe is sufficiently similar to the weight of the duct or pipe itself, as opposed to a large fan or large heat exchanger. If a component is large enough to require support that is independent of the piping,

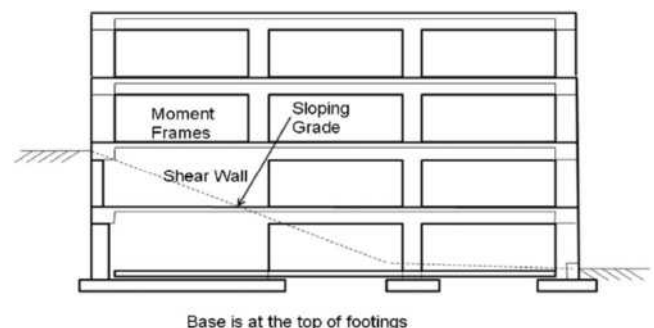


FIGURE C11.2-9 Building with Vertical Elements of the Seismic Force-Resisting System Supporting Lateral Earth Pressures

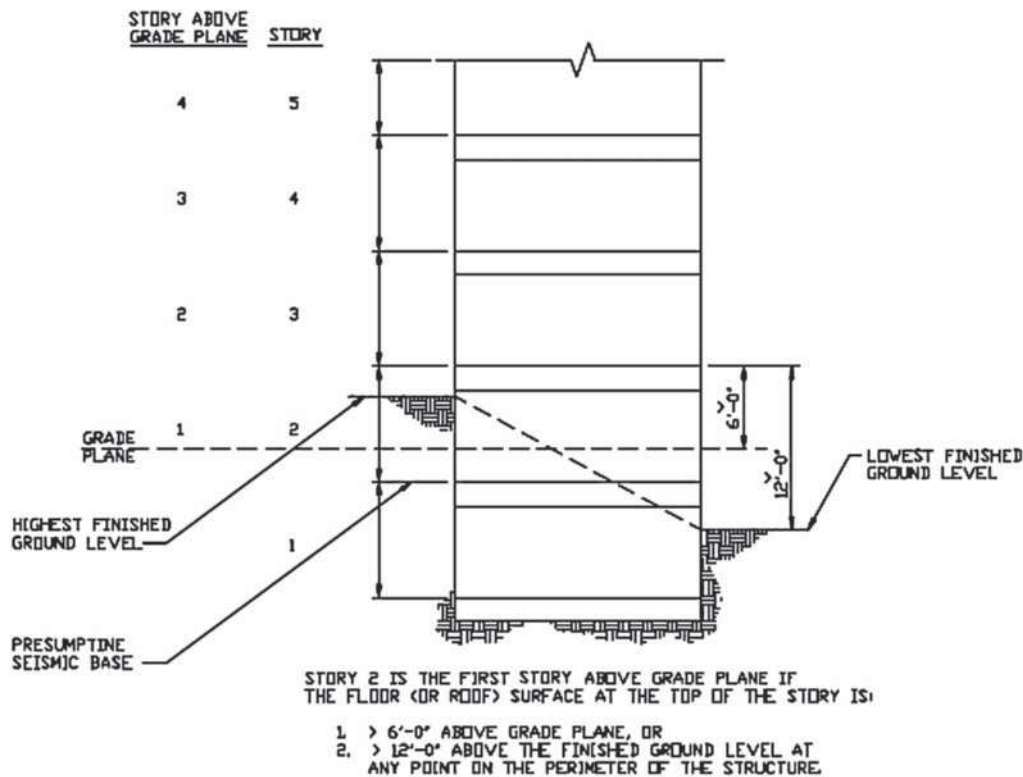


FIGURE C11.2-10 Illustration of Definition of Story above Grade Plane

duct, or conduit to which it is attached, it should likely be treated as a discrete component with regard to both exemptions and general design requirements. Representative distribution systems are listed in Table 13.6-1.

FLEXURE-CONTROLLED DIAPHRAGM: An example of a flexure-controlled diaphragm is a cast-in-place concrete diaphragm, where the flexural yielding mechanism would typically be yielding of the chord tension reinforcement.

SHEAR-CONTROLLED DIAPHRAGM: Shear-controlled diaphragms fall into two main categories. The first category is diaphragms that cannot develop a flexural mechanism because of aspect ratio, chord member strength, or other constraints. The second category is diaphragms that are intended to yield in shear rather than in flexure. Wood-sheathed diaphragms, for example, typically fall in the second category.

STORY ABOVE GRADE PLANE: Fig. C11.2-10 illustrates this definition.

TRANSFER FORCES: Transfer forces are diaphragm forces that are not caused by the acceleration of the diaphragm inertial mass. Transfer forces occur because of discontinuities in the vertical elements of the seismic force-resisting system or because of changes in stiffness in these vertical elements from one story to the next, even if there is no discontinuity. Additionally, buildings that combine frames and shear walls, which would have different deflected shapes under the same loading, also develop transfer forces in the diaphragms that constrain the frames and shear walls to deform together; this development is especially significant in dual systems.

C11.3 SYMBOLS

The provisions for precast concrete diaphragm design are intended to ensure that yielding, when it occurs, is ductile. Since yielding in shear is generally brittle at precast concrete

connections, an additional overstrength factor, Ω_v , has been introduced; the required shear strength for a precast diaphragm is required to be amplified by this factor. This term is added to the symbols.

δ_{MDD} = This symbol refers to in-plane diaphragm deflection and is therefore designated with a lower-case delta. Note that the definition for δ_{MDD} refers to "lateral load" without any qualification, and the definition for Δ_{ADVE} refers to "tributary lateral load equivalent to that used in the computation of δ_{MDD} ." This equivalency is an important concept that was part of the 1997 Uniform Building Code (UBC) (ICBO 1997) definition for a flexible diaphragm.

Ω_v = The provisions for precast concrete diaphragm design are intended to ensure that yielding, when it occurs, is ductile. Since yielding in shear is generally brittle at precast concrete connections, an additional overstrength factor, Ω_v , has been introduced; the required shear strength for a precast diaphragm is required to be amplified by this factor. This term is added to the symbols.

C11.4 SEISMIC GROUND MOTION VALUES

The basis for the mapped values of the MCE_R ground motions in ASCE 7-16 is identical to that in ASCE 7-10. Both of these are significantly different from mapped values of MCE ground motions in earlier editions of ASCE 7. These differences include use of (1) probabilistic ground motions that are based on uniform risk, rather than uniform hazard, (2) deterministic ground motions that are based on the 84th percentile (approximately 1.8 times median), rather than 1.5 times median response spectral acceleration for sites near active faults, and (3) ground motion intensity that is based on maximum rather than average

(geometric mean) response spectral acceleration in the horizontal plane. These differences are explained in detail in the Commentary of the 2009 *NEHRP Recommended Provisions*. Except for determining the MCE_G PGA values in Chapters 11 and 21, the mapped values are given as MCE_R spectral values.

C11.4.1 Near-Fault Sites. In addition to very large accelerations, ground motions on sites located close to the zone of fault rupture of large-magnitude earthquakes can exhibit impulsive characteristics as well as unique directionality not typically recorded at sites located more distant from the zone of rupture. In past earthquakes, these characteristics have been observed to be particularly destructive. Accordingly, this standard establishes more restrictive design criteria for structures located on sites where such ground motions may occur. The standard also requires direct consideration of these unique characteristics in selection and scaling of ground motions used in nonlinear response history analysis and for the design of structures using seismic isolation or energy-dissipation devices when located on such sites.

The distance from the zone of fault rupture at which these effects can be experienced is dependent on a number of factors, including the rupture type, depth of fault, magnitude, and direction of fault rupture. Therefore, a precise definition of what constitutes a near-fault site is difficult to establish on a general basis. This standard uses two categorizations of near-fault conditions, both based on the distance of a site from a known active fault, capable of producing earthquakes of a defined magnitude or greater, and having average annual slip rates of nonnegligible amounts. These definitions were first introduced in the 1997 UBC (ICBO 1997). Fig. C11.4-1 illustrates the means of determining the distance of a site from a fault, where the fault plane dips at an angle relative to the ground surface.

C11.4.2 Mapped Acceleration Parameters. Mapped response spectral accelerations (5% damping) are provided on U.S. Geological Survey (USGS) maps for short periods, S_S , and at 1 s, S_1 , for sites at the boundary of Site Classes B and C, which is $\bar{v}_s = 760$ m/s (2,500 ft/s). The USGS maps have been applicable to this site condition since 1996 (Frankel et al. 1996), but now are more clearly marked as being applicable to the reference value of $\bar{v}_s$. USGS ground motion maps are available at <https://doi.org/10.5066/F7NK3C76>.

C11.4.3 Site Class. The new site coefficients, F_a and F_v , necessitated a revision to the default site class when the site is

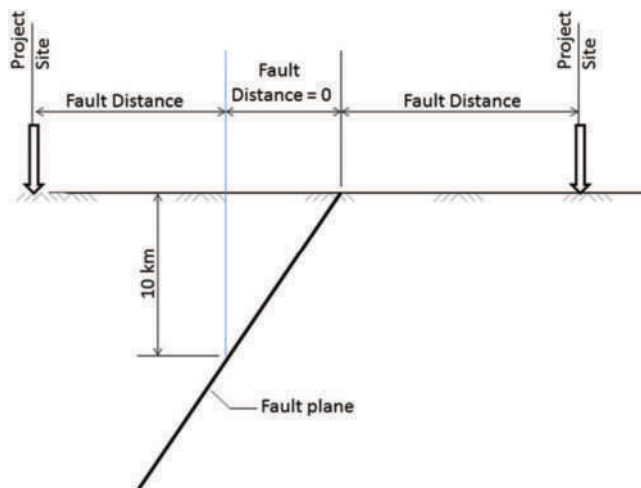


FIGURE C11.4-1 Fault Distance for Various Project Site Locations

known to be soil not classified as Site Class E or F. The F_a and F_v values for Site Class D in ASCE 7-10 were always equal to or greater than F_a and F_v values for Site Class C. Thus, specifying Site Class D as the default site class ensured that the response spectral accelerations would not be underestimated. However, the F_a values for Site Class C are greater than those for Site Class D for $S_S \geq 1.0$. Thus, a minor modification was required, consisting of adding the sentence “Where Site Class D is selected as the default site class per Section 11.4.3, the value of F_a shall not be less than 1.2.” in the last paragraph of Section 11.4.4 to ensure that the larger of the site coefficients for Site Classes C and D would be selected when the soil properties are not sufficiently known to determine the site class, and the authority having jurisdiction or geotechnical data have determined that Site Class E or F soils are not present.

Because the site coefficients, F_a and F_v , are less than unity (1.0) for Site Class B, a new paragraph was added to this section that requires the measurement of shear-wave velocity to demonstrate that the site is Site Class B according to the definition in Section 20.3. Furthermore, when $S_1 \geq 1.0$, the values of F_a for Site Class C are now greater than those for Site Class D.

C11.4.4 Site Coefficients and Risk-Targeted Maximum Considered Earthquake (MCE_R) Spectral Response Acceleration Parameters. Acceleration response parameters obtained from the maps (figures) cited in Section 11.4.2 are applicable for sites that have $\bar{v}_s = 760$ m/s ($\bar{v}_s = 2,500$ ft/s). For other site conditions, the S_S and S_1 values are computed as indicated in Section 11.4.4. This section has been revised from ASCE 7-10 to adjust the site factors to a reference site condition of $\bar{v}_s = 760$ m/s ($\bar{v}_s = 2,500$ ft/s) (instead of Site Class B) and to reflect more recent knowledge and data pertaining to site response.

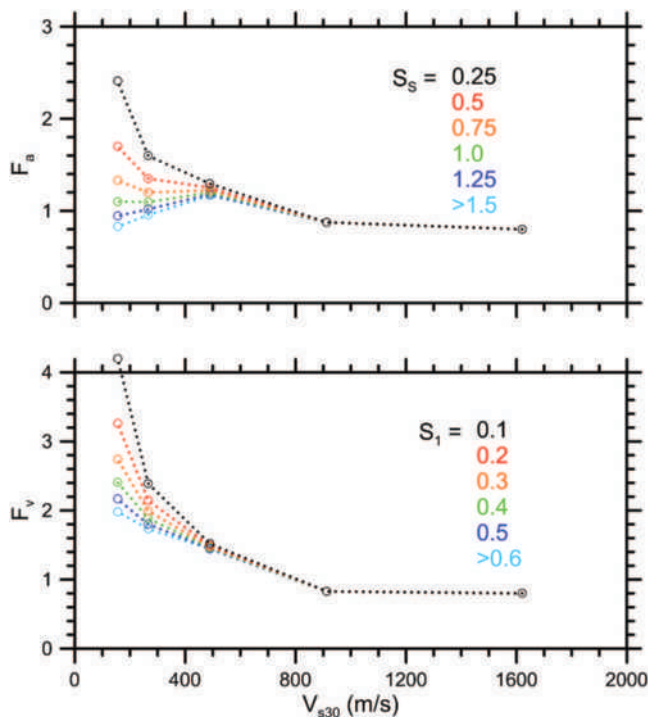
The site coefficients, F_a and F_v , presented, respectively, in Tables 11.4-1 and 11.4-2 for the various site classes are based on analysis of strong-motion data and on numerical simulations of nonlinear site response. The development of the factors that were in place from the 1994 *NEHRP Provisions* through ASCE 7-10 is described in Dobry et al. (2000) and the references therein. Motivation for the revisions to these site factors includes the following (Seyhan and Stewart 2012): (1) updating the reference site condition used for the factors to match the condition on the national maps, which is $\bar{v}_s = 760$ m/s ($\bar{v}_s = 2,500$ ft/s); and (2) incorporating into the factors the substantial knowledge gains (stemming in large part from an enormous increase in available data) on site response over the past two decades.

The work undertaken to develop the revised factors is described in two PEER reports (Boore et al. 2013, Stewart and Seyhan 2013). They develop a semiempirical site amplification model for shallow crustal regions with two components: (1) a component to account for the change in ground motion with $\bar{v}_s$ for weak shaking conditions (referred to as the $\bar{v}_s$ -scaling component); and (2) a component accounting for the effect of nonlinearity. The $\bar{v}_s$ -scaling component was derived from strong ground motion data compiled and analyzed as part of the NGA-West 2 project (<http://peer.berkeley.edu/ngawest2/>). Whereas Stewart and Seyhan (2013) describe some regional variations in $\bar{v}_s$ -scaling, such variations were not judged to be sufficiently robust for inclusion in the model, and therefore the model's $\bar{v}_s$ -scaling reflects the average of the full international data set. The nonlinear component of the model is designed to jointly capture nonlinear effects revealed by strong-motion data analysis and the results of numerical simulations by Kamai et al. (2013).

The complete model (incorporating both $\bar{v}_s$ -scaling and nonlinearity) is used to derive the recommended values of F_a and F_v for Site Classes B–D. The reference velocity used in the

computations was 760 m/s (2,500 ft/s). The values of $\bar{v}_s$ used to compute the tabulated factors for Classes B, C, and D were 913 m/s (2,995 ft/s), 489 m/s (1,604 ft/s), and 266 m/s (873 ft/s), respectively. These are average values of $\bar{v}_s$ for sites in the respective classes based on the NGA-West 2 data set. For Site Class E, median estimates of site amplification were computed using the complete model (applied at 155 m/s (509 ft/s)) as with the other classes. However, the recommended factors for Site Class E are increased above the median by half of the within-event standard deviation derived from the data, which increases site factors by approximately a factor of 1.3 to 1.4. This increase introduces a conservative bias to the Class E factors that is considered desirable because of the relatively modest amount of data for this site condition. A conservative bias was applied in the original site factors for Class E as well.

Fig. C11.4-2 shows the recommended site factors as a function of $\bar{v}_s$ for the levels of excitation (specified as values of S_s and S_1) given in Tables 11.4-1 and 11.4-2. The revised site factors for Site Class B (rock) are smaller than earlier values because of the change in reference velocity from 1,050 to 760 m/s (3,445 to 2,500 ft/s). The revised values for soil sites are generally similar to the prior values. However, for stronger shaking levels and Site Class C and D soils, the revised site factors are the same as or greater than the earlier values because of reduced levels of nonlinearity, especially at long periods (i.e., in the F_v parameter). For Class E, the relative levels of revised and prior site factors are strongly influenced by the amount of conservative bias adopted in their selection.



Note: In the top frame of this figure, the values of F_a for $V_{s30} = 155$ m/s (Site Class E) are 1.1, 1.0, and 0.8 for $S_s = 1.0$, 1.25, and >1.5 , respectively. These values were originally approved by the Provisions Update Committee but were later replaced by the note, "See Sec. 11.4.8," which was necessitated by the results of more recent research. See Section C11.4.8 for details.

FIGURE C11.4-2 Site Factors F_a and F_v as Function of $\bar{v}_s$ (Shown as V_{s30} in Figure) for Various Amplitudes of Reference Rock Shaking.

Source: Stewart and Seyhan 2013.

Whereas the overall levels of Class E site amplification remain about the same, the degree of nonlinearity has been reduced somewhat for F_a and increased somewhat for F_v .

The revised factors are applicable for average site conditions in tectonically active regions (e.g., the West Coast of the United States). Because of different average site conditions in stable continental regions (such as the central and eastern United States), differences in average site response relative to the factors in Tables 11.4-1 and 11.4-2 should be anticipated for such regions. This difference can be addressed through site-specific analysis performed in accordance with Section 11.4.8.

For Site Classes B–D, site coefficients F_a and F_v may be computed from the following equations in lieu of using the site factors in Tables 11.4-1 and 11.4-2:

$$F_a = \exp \left[-0.727 \ln \left(\frac{V_{s30}}{760} \right) - 0.2298 \left[\frac{\exp \{ -0.00638 (\min(V_{s30}, 760) - 360) \}}{\exp \{ -0.00638 \times 400 \}} \right] \times \ln \left(\frac{(S_s/2.3) + 0.1}{0.1} \right) \right] \quad (\text{C11.4-1})$$

$$F_v = \exp \left[-1.03 \ln \left(\frac{V_{s30}}{760} \right) - 0.118 \left[\frac{\exp \{ -0.00756 (\min(V_{s30}, 760) - 360) \}}{\exp \{ -0.00756 \times 400 \}} \right] \times \ln \left(\frac{(S_1/0.7) + 0.1}{0.1} \right) \right] \quad (\text{C11.4-2})$$

In Eqs. (C11.4-1) and (C11.4-2), $\bar{v}_s$ (shown as V_{s30} in the equations) is in units of m/s and S_s and S_1 are in units of g. The equations are considered useful for $\bar{v}_s = 150$ to 1,000 m/s ($\bar{v}_s = 492$ to 3,281 ft/s), $S_s = 0 - 1.8$ g, and $S_1 = 0 - 0.6$ g (gravity). To obtain the F_a and F_v for $\bar{v}_s < 180$ m/s ($\bar{v}_s < 590$ ft/s), the $+1/2$ standard-deviation correction for Site Class E described earlier in this section would need to be applied to the natural logarithm of F_a and F_v resulting from both equations. The standard deviations are 0.67 for F_a and 0.58 for F_v . One half of these standard deviations are to be added to the natural logarithms of F_a and F_v ; the antilogs of the resulting values yield F_a and F_v for Site Class E. Equations applicable for specific periods, and that allow use of $\bar{v}_s > 1,000$ m/s ($\bar{v}_s > 3,281$ ft/s), are given in Boore et al. (2013).

C11.4.5 Design Spectral Acceleration Parameters. As described in Section C11.4, structural design in ASCE 7 is performed for earthquake demands that are 2/3 of the MCE_R response spectra. As set forth in Section 11.4.5, two additional parameters, S_{DS} and S_{D1} , are used to define the acceleration response spectrum for this design level event. These parameters are 2/3 of the respective S_{MS} and S_{M1} values and define a design response spectrum for sites of any characteristics and for natural periods of vibration less than the transition period, T_L . Values of S_{MS} , S_{M1} , S_{DS} , and S_{D1} can also be obtained from the USGS website cited previously.

C11.4.6 Design Response Spectrum. The design response spectrum (Fig. 11.4-1) consists of several segments. The constant-acceleration segment covers the period band from T_0 to T_s ; response accelerations in this band are constant and equal to S_{DS} . The constant-velocity segment covers the period band

from T_s to T_L , and the response accelerations in this band are proportional to $1/T$ with the response acceleration at a 1-s period equal to S_{D1} . The long-period portion of the design response spectrum is defined on the basis of the parameter, T_L , the period that marks the transition from the constant-velocity segment to the constant-displacement segment of the design response spectrum. Response accelerations in the constant-displacement segment, where $T \geq T_L$, are proportional to $1/T^2$. Values of T_L are provided on maps in Figs. 22-14 through 22-17.

The T_L maps were prepared following a two-step procedure. First, a correlation between earthquake magnitude and T_L was established. Then, the modal magnitude from deaggregation of the ground-motion seismic hazard at a 2-s period (a 1-s period for Hawaii) was mapped. Details of the procedure and the rationale for it are found in Crouse et al. (2006).

C11.4.8 Site-Specific Ground Motion Procedures. Site-specific ground motions are permitted for design of any structure and are required for design of certain structures and certain site soil conditions. The objective of a site-specific ground motion analysis is to determine ground motions for local seismic and site conditions with higher confidence than is possible using the general procedure of Section 11.4.

As noted earlier, the site-specific procedures of Chapter 21 are the same as those used by the U.S. Geological Survey to develop the mapped values of MCE_R ground motions shown in Figs. 22-1 through 22-8 of Chapter 22. Unless significant differences in local seismic and site conditions are determined by a site-specific analysis of earthquake hazard, site-specific ground motions would not be expected to differ significantly from those of the mapped values of MCE_R ground motions prepared by the USGS.

Site-specific ground motions are required for design of structures at softer soil sites and stronger ground motion intensities for which the two domains of constant acceleration and constant velocity (e.g., of the design response spectrum) do not adequately characterize site response and MCE_R response spectral acceleration cannot be reliably calculated using procedures and formulas of Section 11.4. Softer soil sites requiring site-specific ground motions were identified by a study that investigated and developed solutions to potential shortcomings in equivalent lateral force (ELF) (and modal response spectrum analysis, MRSA) design procedures (Kircher & Associates 2015). The impetus for the ELF study came from a BSSC Provisions Update Committee effort (late in the 2015 cycle) to define seismic design forces at additional response periods beyond 1.0 s; a first step toward ultimately basing seismic design forces on multiperiod MCE_R response spectra.

Multiperiod MCE_R response spectra would eliminate potential shortcomings associated with the use of seismic forces based on only two response periods by directly providing reliable values of seismic demand at all design periods of interest. Unfortunately, multiperiod hazard and associated design methods are not yet mature enough for incorporation in seismic codes, and the site-specific requirements of Section 11.4.8 for softer sites and stronger ground motions provide a short-term solution to a problem that will ultimately be resolved by adoption of design methods based on multiperiod response spectra.

The value of parameter S_{MS} is based on response at a period of 0.2 s, and the value of the parameter S_{M1} is based on response at a period of 1 s. The domain of constant acceleration defined by the parameter (S_{MS}) and the domain of constant velocity (S_{M1}/T) are crude approximations to the actual shape of response spectral accelerations of MCE_R ground motions, such as those calculated using the site-specific procedures of Chapter 21 for a number of different periods of response (so-called multiperiod MCE_R response spectra).

Although approximate, the two domains of constant acceleration and velocity provide reasonably accurate and conservative representation of the frequency content of design ground motions when peak response spectral acceleration occurs at or near $T = 0.2$ s, the period used to define S_{MS} , and peak response spectral velocity (i.e., peak response spectral acceleration divided by response period) occurs at or near $T = 1.0$ s, the period used to define S_{M1} . Such is the case for response at stiffer sites governed by smaller magnitude earthquakes, but generally it is inaccurate and potentially unconservative at softer sites (e.g., Site Classes D and E), in particular sites for which seismic hazard is dominated by large-magnitude earthquakes. In the latter case, values of S_{MS} and S_{M1} would be more accurately calculated if based on response at periods that better represent peak response spectral acceleration and peak response spectral velocity, and hence the frequency content, of MCE_R ground motions of the site of interest.

The site-specific requirements of Section 21.4 of ASCE 7-10 recognized that periods of peak response are not always at 0.2 s and 1.0 s and require that S_{D1} be taken as equal to 2 times the response spectral acceleration at 2 s with greater than 1-s response spectral acceleration, and that S_{D5} be taken as equal to 0.2-s response spectral acceleration, but not less than 90% of response at any period to better represent the frequency content of site-specific ground motions. For softer sites governed by large-magnitude events, the peak value of response spectral velocity can occur at a period beyond 2 s, and ASCE 7-16 extends the S_{D1} criteria of Section 21.4 to a period range of 1 s to 5 s for Site Class D and E sites (i.e., $V_{s30} \leq 1,200$ ft/s [$V_{s30} \leq 365.76$ m/s]).

Potential shortcomings in ELF seismic design forces are illustrated in Figs. C11.4-3, C11.4-4, and C11.4-5, each of which shows plots of a multiperiod MCE_R response spectrum for Site Class BC, multiperiod MCE_R , design response spectra for the site class of interest (Site Class C, D, or E), and the two-domain ELF design spectrum, defined by the product $C_s \times (R/I_e)$. In these figures, the MCE_R ground motions represent a magnitude 7.0 earthquake at $R_x = 6.75$ km ($R_x = 4.19$ mi), which has values of the parameters $S_S = 1.5$ g and $S_1 = 0.6$ g for Site Class BC conditions [$V_s V_{s30} = 2,500$ ft/s ($V_s V_{s30} = 762$ m/s)]. The ELF design spectrum is based on these values of S_S and S_1 and values of the site coefficients F_a and F_v for the site class of

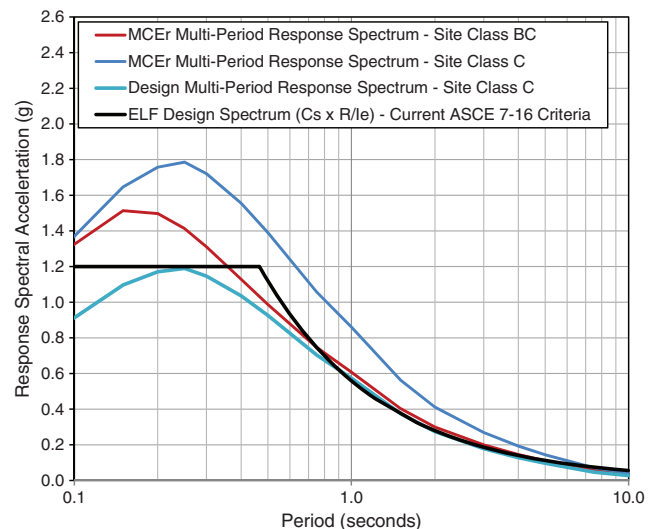


FIGURE C11.4-3 Comparison of ELF and Multiperiod Design Spectra—Site Class C Ground Motions ($V_s V_{s30} = 1,600$ ft/s)

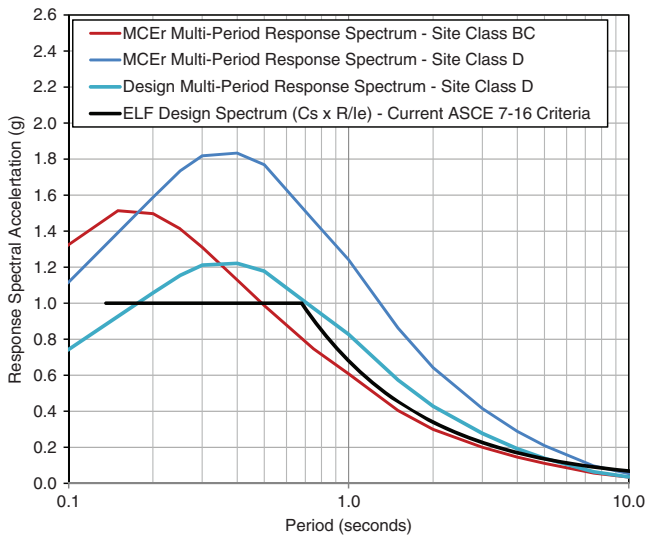


FIGURE C11.4-4 Comparison of ELF and Multiperiod Design Spectra—Site Class D Ground Motions ($V_{s30} = 870$ ft/s)

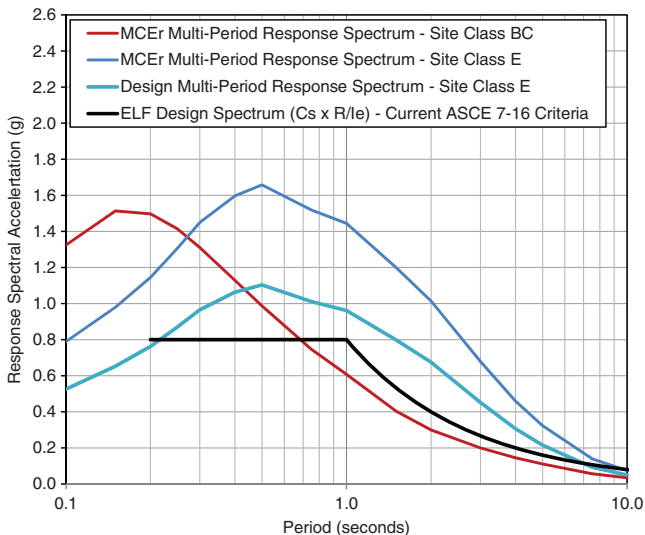


FIGURE C11.4-5 Comparison of ELF and Multiperiod Design Spectra—Site Class E Ground Motions ($V_s v_{s30} = 510$ ft/s)

interest. For example, the domain of constant acceleration is defined by the value of the parameter $S_{DS} = 2/3 \times 0.8 \times 1.5 \text{ g} = 0.8 \text{ g}$, and the domain of constant velocity is defined by the value of the parameter $S_{D1} = 2/3 \times 2.0 \times 0.6 \text{ g} = 0.8 \text{ g}$ for the ELF design spectrum shown in Fig. C11.4-5 for Site Class E conditions.

Comparisons of multiperiod and ELF design spectra in Figs. C11.4-3, C11.4-4, and C11.4-5 show varying degrees of similarity. For Site Class C (Fig. C11.4-3), the ELF design spectrum is similar to the multiperiod design spectrum. The ELF domain of constant acceleration matches the peak of the multiperiod design spectrum at period of 0.25 s, and the ELF domain of constant velocity ($1/T$) matches the multiperiod design spectrum at periods of 1 s and greater. For Site Class D (Fig. C11.4-4), the ELF design spectrum is moderately unconservative at most periods (e.g., by about 20% at a period of 1 s to 2 s), and for Site Class E (Fig. C11.4-5), the design spectrum is

significantly unconservative at all periods (e.g., by about a factor of 1.65 at periods of 2 s to 3 s). These figures are based on multiperiod response spectra whose shape corresponds to a magnitude 7.0 earthquake. Spectral shape is a function of magnitude, and smaller magnitude events would show greater similarity between the multiperiod design spectrum and ELF spectrum of site class of interest, whereas larger magnitude events would show more significant differences.

In general, Section 11.4.8 requires site-specific hazard analysis for structures on Site Class E with values of S_S greater than or equal to 1.0 g, and for structures on Site Class D or Site Class E for values of S_1 greater than or equal to 0.2 g. These requirements significantly limit the use of practical ELF and MRSA design methods, which is of particular significance for Site Class D sites. To lessen the effect of these requirements on design practice, three exceptions permit the use of conservative values of design parameters for certain conditions for which conservative values of design were identified by the ELF study. These exceptions do not apply to seismically isolated structures and structures with damping systems for which site-specific analysis is required in all cases at sites with S_1 greater than or equal to 0.6.

The first exception permits use of the value of the site coefficient F_a of Site Class C ($F_a = 1.2$) for Site Class E sites (for values of S_S greater than or equal to 1.0 g) in lieu of site-specific hazard analysis. The ELF study found that while values of the site coefficient F_a tend to decrease with intensity for softer sites, values of spectrum shape adjustment factor C_a tend to increase such that the net effect is approximately the same intensity of MCE_R ground motions for Site Classes C, D, and E when MCE_R ground motion intensity is strong (i.e., $S_{MS} \geq 1.0$). Site Class C was found to not require spectrum shape adjustment, and the value of site coefficient F_a for Site Class C ($F_a = 1.2$) is large enough to represent both site class and spectrum shape effects for Site Class E (and Site Class D).

The second exception permits both ELF and MRSA design of structures at Site Class D sites for values of S_1 greater than or equal to 0.2 g, provided that the value of the seismic response coefficient C_s is conservatively calculated using Eq. (12.8-2) for $T \leq 1.5T_s$ and using 1.5 times the value computed in accordance with either Eq. (12.8-3) for $T_L \geq T > 1.5T_s$ or Eq. (12.8-4) for $T > T_L$. This exception recognizes that structures are conservatively designed for the response spectral acceleration defined by the domain of constant acceleration (S_{DS}) or by a 50% increase in the value of seismic response coefficient C_s for structures with longer periods ($T \geq 1.5T_s$). The underlying presumption of this exception for MRSA design of structures is that the shape of the design response spectrum (Fig. 11.4-1) is sufficiently representative of the frequency content of Site Class D ground motions to permit use of MRSA and that the potential underestimation of fundamental-mode response using the design response spectrum shape of Fig. 11.4-1 is accounted for by scaling MRSA design values (Section 12.9.1.4) with a conservative value of the seismic response coefficient C_s . In general, this exception effectively limits the requirements for site-specific hazard analysis to very tall and or flexible structures at Site Class D sites ($S_1 \geq 0.2 \text{ g}$).

The third exception permits ELF design of short-period structures ($T \leq T_s$) at Site Class E sites for values of S_S greater than or equal to 0.2 g. This exception recognizes that short-period structures are conservatively designed using the ELF procedure for values of seismic response coefficient C_s based on the domain of constant acceleration (S_{DS}), which is, in all cases, greater than or equal to response spectral accelerations of the domain of constant velocity and therefore need not consider the effects of spectrum shape at periods $T > T_s$. In general, the shape of the design response spectrum (Fig. 11.4-1) is not representative of the frequency content of Site Class E ground motions, and

MRSA is not permitted for design unless the design spectrum is calculated using the site-specific procedures of Section 21.2.

C11.5 IMPORTANCE FACTOR AND RISK CATEGORY

Large earthquakes are rare events that include severe ground motions. Such events are expected to result in damage to structures even if they were designed and built in accordance with the minimum requirements of the standard. The consequence of structural damage or failure is not the same for the various types of structures located within a given community. Serious damage to certain classes of structures, such as critical facilities (e.g., hospitals), disproportionately affects a community. The fundamental purpose of this section and of subsequent requirements that depend on this section is to improve the ability of a community to recover from a damaging earthquake by tailoring the seismic protection requirements to the relative importance of a structure. That purpose is achieved by requiring improved performance for structures that

1. Are necessary to response and recovery efforts immediately after an earthquake,
2. Present the potential for catastrophic loss in the event of an earthquake, or
3. House a large number of occupants or occupants less able to care for themselves than the average.

The first basis for seismic design in the standard is that structures should have a suitably low likelihood of collapse in the rare events defined as the maximum considered earthquake (MCE) ground motion. A second basis is that life-threatening damage, primarily from failure of nonstructural components in and on structures, is unlikely in a design earthquake ground motion (defined as two-thirds of the MCE). Given the occurrence of ground motion equivalent to the MCE, a population of structures built to meet these design objectives probably still experiences substantial damage in many structures, rendering these structures unfit for occupancy or use. Experience in past earthquakes around the world has demonstrated that there is an immediate need to treat injured people, to extinguish fires and prevent conflagration, to rescue people from severely damaged or collapsed structures, and to provide shelter and sustenance to a population deprived of its normal means. These needs are best met when structures essential to response and recovery activities remain functional.

This standard addresses these objectives by requiring that each structure be assigned to one of the four Risk Categories presented in Chapter 1 and by assigning an Importance Factor, I_e , to the structure based on that Risk Category. (The two lowest categories, I and II, are combined for all purposes within the seismic provisions.) The Risk Category is then used as one of two components in determining the Seismic Design Category (see Section C11.6) and is a primary factor in setting drift limits for building structures under the design earthquake ground motion (see Section C12.12).

Fig. C11.5-1 shows the combined intent of these requirements for design. The vertical scale is the likelihood of the ground motion; the MCE is the rarest considered. The horizontal scale is the level of performance intended for the structure and attached nonstructural components, which range from collapse to operational. The basic objective of collapse prevention at the MCE for ordinary structures (Risk Category II) is shown at the lower right by the solid triangle; protection from life-threatening damage at the design earthquake ground motion (defined by the standard as two-thirds of the MCE) is shown by the hatched triangle. The performance implied for higher Risk Categories III and IV is shown by squares and circles, respectively. The performance anticipated for less severe ground motion is shown by open symbols.

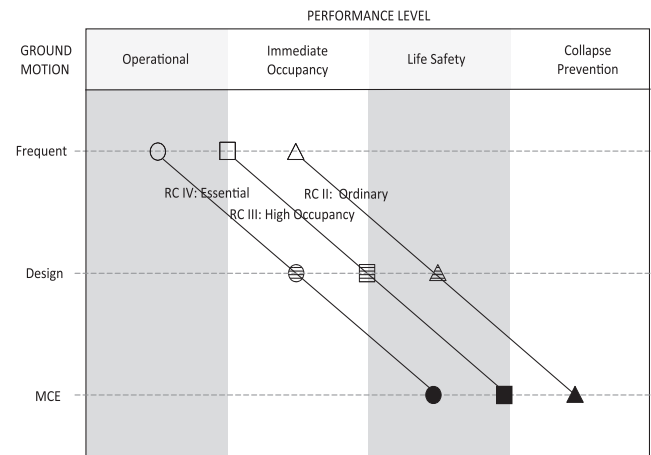


FIGURE C11.5-1 Expected Performance as Related to Risk Category and Level of Ground Motion

C11.5.1 Importance Factor. The Importance Factor, I_e , is used throughout the standard in quantitative criteria for strength. In most of those quantitative criteria, the Importance Factor is shown as a divisor on the factor R or R_p to reduce damage for important structures in addition to preventing collapse in larger ground motions. The R and R_p factors adjust the computed linear elastic response to a value appropriate for design; in many structures, the largest component of that adjustment is ductility (the ability of the structure to undergo repeated cycles of inelastic strain in opposing directions). For a given strength demand, reducing the effective R factor (by means of the Importance Factor) increases the required yield strength, thus reducing ductility demand and related damage.

C11.5.2 Protected Access for Risk Category IV. Those structures considered essential facilities for response and recovery efforts must be accessible to carry out their purpose. For example, if the collapse of a simple canopy at a hospital could block ambulances from the emergency room admittance area, then the canopy must meet the same structural standard as the hospital. The protected access requirement must be considered in the siting of essential facilities in densely built urban areas.

C11.6 SEISMIC DESIGN CATEGORY

Seismic Design Categories (SDCs) provide a means to step progressively from simple, easily performed design and construction procedures and minimums to more sophisticated, detailed, and costly requirements as both the level of seismic hazard and the consequence of failure escalate. The SDCs are used to trigger requirements that are not scalable; such requirements are either on or off. For example, the basic amplitude of ground motion for design is scalable—the quantity simply increases in a continuous fashion as one moves from a low hazard area to a high hazard area. However, a requirement to avoid weak stories is not particularly scalable. Requirements such as this create step functions. There are many such requirements in the standard, and the SDCs are used systematically to group these step functions. (Further examples include whether seismic anchorage of nonstructural components is required or not, whether particular inspections will be required or not, and structural height limits applied to various seismic force-resisting systems.)

In this regard, SDCs perform one of the functions of the seismic zones used in earlier U.S. building. However, SDCs also depend on a building's occupancy and, therefore, its desired performance. Furthermore, unlike the traditional implementation

of seismic zones, the ground motions used to define the SDCs include the effects of individual site conditions on probable ground-shaking intensity.

In developing the ground-motion limits and design requirements for the various Seismic Design Categories, the equivalent modified Mercalli intensity (MMI) scale was considered. There are now correlations of the qualitative MMI scale with quantitative characterizations of ground motions. The reader is encouraged to consult any of a great many sources that describe the MMIs. The following list is a coarse generalization:

- MMI V No real damage
- MMI VI Light nonstructural damage
- MMI VII Hazardous nonstructural damage
- MMI VIII Hazardous damage to susceptible structures
- MMI IX Hazardous damage to robust structures

When the current design philosophy was adopted from the 1997 NEHRP provisions and Commentary (FEMA 1997a and FEMA 1997b), the upper limit for SDC A was set at roughly one-half of the lower threshold for MMI VII, and the lower limit for SDC D was set at roughly the lower threshold for MMI VIII. However, the lower limit for SDC D was more consciously established by equating that design value (two-thirds of the MCE) to one-half of what had been the maximum design value in building codes over the period of 1975 to 1995. As more correlations between MMI and numerical representations of ground motion have been created, it is reasonable to make the following correlation between the MMI at MCE ground motion and the Seismic Design Category (all this discussion is for ordinary occupancies):

- MMI V SDC A
- MMI VI SDC B
- MMI VII SDC C
- MMI VIII SDC D
- MMI IX SDC E

An important change was made to the determination of SDC when the current design philosophy was adopted. Earlier editions of the *NEHRP Provisions* used the peak velocity-related acceleration, A_v , to determine a building's seismic performance category. However, this coefficient does not adequately represent the damage potential of earthquakes on sites with soil conditions other than rock. Consequently, the 1997 NEHRP provisions (FEMA 1997a) adopted the use of response spectral acceleration parameters S_{DS} and S_{D1} , which include site soil effects for this purpose.

Except for the lowest level of hazard (SDC A), the SDC also depends on the Risk Categories. For a given level of ground motion, the SDC is one category higher for Risk Category IV structures than for lower risk structures. This rating has the effect of increasing the confidence that the design and construction requirements can deliver the intended performance in the extreme event.

Note that the tables in the standard are at the design level, defined as two-thirds of the MCE level. Also recall that the MMIs are qualitative by their nature and that the above correlation will be more or less valid, depending on which numerical correlation for MMI is used. The numerical correlations for MMI roughly double with each step, so correlation between design earthquake ground motion and MMI is not as simple or convenient.

In sum, at the MCE level, SDC A structures should not see motions that are normally destructive to structural systems, whereas the MCE level motions for SDC D structures can destroy vulnerable structures. The grouping of step function requirements by SDC is such that there are a few basic structural integrity requirements imposed at SDC A, graduating to a suite of requirements at SDC D based on observed performance in past earthquakes, analysis, and laboratory research.

The nature of ground motions within a few kilometers of a fault can be different from more distant motions. For example, some near-fault motions have strong velocity pulses, associated with forward rupture directivity, that tend to be highly destructive to irregular structures, even if they are well detailed. For ordinary occupancies, the boundary between SDCs D and E is set to define sites likely to be close enough to a fault that these unusual ground motions may be present. Note that this boundary is defined in terms of mapped bedrock outcrop motions affecting response at 1 s, not site-adjusted values, to better discriminate between sites near and far from faults. Short-period response is not normally as affected as the longer period response. The additional design criteria imposed on structures in SDCs E and F specifically are intended to provide acceptable performance under these very intense near-fault ground motions.

For most buildings, the SDC is determined without consideration of the building's period. Structures are assigned to an SDC based on the more severe condition determined from 1-s acceleration and short-period acceleration. This assigning is done for several reasons. Perhaps the most important of these is that it is often difficult to estimate precisely the period of a structure using default procedures contained in the standard. Consider, for example, the case of rigid wall-flexible diaphragm buildings, including low-rise reinforced masonry and concrete tilt-up buildings with either untopped metal deck or wood diaphragms. The formula in the standard for determining the period of vibration of such buildings is based solely on the structural height, h_n , and the length of wall present. These formulas typically indicate very short periods for such structures, often on the order of 0.2 s or less. However, the actual dynamic behavior of these buildings often is dominated by the flexibility of the diaphragm—a factor neglected by the formula for approximate fundamental period. Large buildings of this type can have actual periods on the order of 1 s or more. To avoid misclassifying a building's SDC by inaccurately estimating the fundamental period, the standard generally requires that the more severe SDC determined on the basis of short- and long-period shaking be used.

Another reason for this requirement is a desire to simplify building regulation by requiring all buildings on a given soil profile in a particular region to be assigned to the same SDC, regardless of the structural type. This assignment has the advantage of permitting uniform regulation in the selection of seismic force-resisting systems, inspection and testing requirements, seismic design requirements for nonstructural components, and similar aspects of the design process regulated on the basis of SDC, within a community.

Notwithstanding the above, it is recognized that classification of a building as SDC C instead of B or D can have a significant impact on the cost of construction. Therefore, the standard includes an exception permitting the classification of buildings that can reliably be classified as having short structural periods on the basis of short-period shaking alone.

Local or regional jurisdictions enforcing building regulations may desire to consider the effect of the maps, typical soil conditions, and Seismic Design Categories on the practices in their jurisdictional areas. For reasons of uniformity of practice or reduction of potential errors, adopting ordinances could stipulate particular values of ground motion, particular site classes, or particular Seismic Design Categories for all or part of the area of their jurisdiction. For example,

1. An area with a historical practice of high seismic zone detailing might mandate a minimum SDC of D regardless of ground motion or site class.

2. A jurisdiction with low variation in ground motion across the area might stipulate particular values of ground motion rather than requiring the use of maps.
3. An area with unusual soils might require use of a particular site class unless a geotechnical investigation proves a better site class.

C11.7 DESIGN REQUIREMENTS FOR SEISMIC DESIGN CATEGORY A

The 2002 edition of the standard included a new provision of minimum lateral force for Seismic Design Category A structures. The minimum load is a structural integrity issue related to the load path. It is intended to specify design forces in excess of wind loads in heavy low-rise construction. The design calculation in Section 1.4.2 of the standard is simple and easily done to ascertain if the seismic load or the wind load governs. This provision requires a minimum lateral force of 1% of the total gravity load assigned to a story to ensure general structural integrity.

Seismic Design Category A is assigned when the MCE ground motions are below those normally associated with hazardous damage. Damaging earthquakes are not unknown or impossible in such regions, however, and ground motions close to such events may be large enough to produce serious damage. Providing a minimum level of resistance reduces both the radius over which the ground motion exceeds structural capacities and resulting damage in such rare events. There are reasons beyond seismic risk for minimum levels of structural integrity.

The requirements for SDC A in Section 1.4 are all minimum strengths for structural elements stated as forces at the level appropriate for direct use in the strength design load combinations of Section 2.3. The two fundamental requirements are a minimum strength for a structural system to resist lateral forces (Section 1.4.2) and a minimum strength for connections of structural members (Section 1.4.3).

For many buildings, the wind force controls the strength of the lateral-force-resisting system, but for low-rise buildings of heavy construction with large plan aspect ratios, the minimum lateral force specified in Section 1.4.2 may control. Note that the requirement is for strength and not for toughness, energy-dissipation capacity, or some measure of ductility. The force level is not tied to any postulated seismic ground motion. The boundary between SDCs A and B is based on a spectral response acceleration of 25% of gravity (MCE level) for short-period structures; clearly the 1% acceleration level (from Eq. (1.4-1)) is far smaller. For ground motions below the A/B boundary, the spectral displacements generally are on the order of a few inches or less depending on period. Experience has shown that even a minimal strength is beneficial in providing resistance to small ground motions, and it is an easy provision to implement in design. The low probability of motions greater than the MCE is a factor in taking the simple approach without requiring details that would produce a ductile response. Another factor is that larger design forces are specified in Section 1.4.3 for connections between main elements of the lateral force load path.

The minimum connection force is specified in three ways: a general minimum horizontal capacity for all connections; a special minimum for horizontal restraint of in-line beams and trusses, which also includes the live load on the member; and a special minimum for horizontal restraint of concrete and masonry walls perpendicular to their plane (Section 1.4.4). The 5% coefficient used for the first two is a simple and convenient value that provides some margin over the minimum strength of the system as a whole.

C11.8 GEOLOGIC HAZARDS AND GEOTECHNICAL INVESTIGATION

In addition to this commentary, Part 3 of the 2009 NEHRP recommended provisions (FEMA 2009) includes additional and more detailed discussion and guidance on evaluation of geologic hazards and determination of seismic lateral pressures.

C11.8.1 Site Limitation for Seismic Design Categories E and F. Because of the difficulty of designing a structure for the direct shearing displacement of fault rupture and the relatively high seismic activity of SDCs E and F, locating a structure on an active fault that has the potential to cause rupture of the ground surface at the structure is prohibited.

C11.8.2 Geotechnical Investigation Report Requirements for Seismic Design Categories C through F. Earthquake motion is only one factor in assessing potential for geologic and seismic hazards. All of the listed hazards can lead to surface ground displacements with potential adverse consequences to structures. Finally, hazard identification alone has little value unless mitigation options are also identified.

C11.8.3 Additional Geotechnical Investigation Report Requirements for Seismic Design Categories D through F.

New provisions for computing peak ground acceleration for soil liquefaction and stability evaluations have been introduced in this section. Of particular note in this section is the explicitly stated requirement that liquefaction must now be evaluated for the MCE_G ground motion. These provisions include maps of the maximum considered earthquake geometric mean (MCE_G) peak ground acceleration (PGA) for Site Class B bedrock plus a site-coefficient (F_{PGA}) table to convert the PGA value to one adjusted for site class effects (PGA_M).

A requirement, similar to the one in Section 11.4.4, has been added to the provisions to take the larger of the F_{PGA} for Site Classes C and D to conservatively account for the amplification when the site is known to consist of soil that is not in Site Class E or F.

The equation used to derive the F_{PGA} values is similar to Eqs. (C11.4-1) and (C11.4-2) for F_a and F_v ; it is as follows:

$$F_{PGA} = \exp \left[-0.604 \ln \left(\frac{\bar{v}_s}{760} \right) - 0.150 \left[\exp \{ -0.00701 (\min(\bar{v}_s, 760) - 360) \} - \exp \{ -0.00701 \times 400 \} \right] \times \ln \left(\frac{PGA + 0.1}{0.1} \right) \right] \quad (C11.8-1)$$

In Eq. (C11.8-1), $\bar{v}_s$ is in units of m/s and PGA is in units of g. Velocities measured in ft/s can be converted to m/s by multiplying by 0.3048. To obtain the F_{PGA} for $\bar{v}_s < 180$ m/s ($\bar{v}_s < 590$ ft/s), the +1/2 standard-deviation correction described for Site Class E in Section C11.4.4 would need to be applied to the natural logarithm of F_{PGA} . The standard deviation is 0.70.

PGA Provisions. Item 2 of Section 11.8.3 states that peak ground acceleration shall be determined based on either a site-specific study, taking into account soil amplification effects, or using Eq. (11.8-1), for which MCE_G peak ground acceleration is obtained from national maps of PGA for bedrock Site Class B multiplied by a site coefficient (F_{PGA}) to obtain peak ground acceleration for other site classes (PGA_M). This methodology for determining peak ground acceleration for liquefaction evaluations improved the methodology in ASCE 7-05 by using mapped

PGA rather than the approximation for PGA by the ratio $S_s/2.5$. Furthermore, in the central and eastern United States, the ratio $S_s/2.5$ tends to underestimate PGA. $S_s/2.5$ is applicable for bedrock Site Class B and thus could be used as input at depth to a site response analysis under the provisions of ASCE 7-05. The use of Eq. (11.8-1) provides an alternative to conducting site response analysis using rock PGA by providing a site-adjusted ground surface acceleration (PGA_M) that can directly be applied in the widely used empirical correlations for assessing liquefaction potential. Correlations for evaluating liquefaction potential are elaborated on in Resource Paper RP 12, "Evaluation of Geologic Hazards and Determination of Seismic Lateral Earth Pressures," published in the 2009 NEHRP provisions (FEMA 2009).

Maps of MCE_G PGA for Site Class B bedrock, similar to maps of S_s and S_1 , are shown in Figs. 22-9 to 22-13 in Chapter 22. Similar to adjustments for the bedrock spectral response accelerations for site response through the F_a and F_v coefficients, bedrock motions for PGA are adjusted for these same site effects using a site coefficient, F_{PGA} , that depends on the level of ground shaking in terms of PGA and the stiffness of the soil, typically defined in terms of the shear-wave velocity in the upper 30 m (98.4 ft) of geologic profile, V_{s30} . Values of F_{PGA} are presented in Table 11.8-1, and the adjustment is made through Eq. (11.8-1), i.e., $PGA_M = F_{PGA} PGA$, where PGA_M is peak ground acceleration adjusted for site class. The method of determining site class, used in the determination of F_a and F_v , is also identical to that in the present and previous ASCE 7 documents.

There is an important difference in the derivation of the PGA maps and the maps of S_s and S_1 in ASCE 7-10. Unlike previous editions of ASCE 7, the S_s and S_1 maps in ASCE 7-10 were derived for the "maximum direction shaking" and are risk based rather than hazard based. However, the PGA maps have been derived based on the geometric mean of the two horizontal components of motion. The geometric mean was used in the PGA maps rather than the PGA for the maximum direction shaking to ensure that there is consistency between the determination of PGA and the basis of the simplified empirical field procedure for estimating liquefaction potential based on results of standard penetration tests (SPTs), cone penetrometer tests (CPTs), and other similar field investigative methods. When these correlations were originally derived, the geomean (or a similar metric) of peak ground acceleration at the ground surface was used to identify the cyclic stress ratio for sites with or without liquefaction. The resulting envelopes of data define the liquefaction cyclic resistance ratio (CRR). Rather than reevaluating these case histories for the "maximum direction shaking," it was decided to develop maps of the geomean PGA and to continue using the existing empirical methods.

Liquefaction Evaluation Requirements. Beginning with ASCE 7-02, it has been the intent that liquefaction potential be evaluated at MCE ground motion levels. There was ambiguity in the previous requirement in ASCE 7-05 as to whether liquefaction potential should be evaluated for the MCE or for the design earthquake. Paragraph 2 of Section 11.8.3 of ASCE 7-05 stated that liquefaction potential would be evaluated for the design earthquake; it also stated that in the absence of a site-specific study, peak ground acceleration shall be assumed to be equal to $S_s/2.5$ (S_s is the MCE short-period response spectral acceleration on Site Class B rock). There has also been a difference in provisions between ASCE 7-05 and the 2006 edition of the IBC, in which Section 1802.2.7 stated that liquefaction shall be evaluated for the design earthquake ground motions and the default value of peak ground acceleration in the absence of a site-specific study was given as $S_{DS}/2.5$ (S_{DS} is the short-period site-adjusted design response spectral acceleration). ASCE 7-10, in

item 2 of Section 11.8.3 and Eq. (11.8-1), requires explicitly that liquefaction potential be evaluated based on the MCE_G peak ground acceleration.

The explicit requirement in ASCE 7-10 to evaluate liquefaction for MCE ground motion rather than to design earthquake ground motion ensures that the full potential for liquefaction is addressed during the evaluation of structure stability, rather than a lesser level when the design earthquake is used. This change also ensures that, for the MCE ground motion, the performance of the structure is considered under a consistent hazard level for the effects of liquefaction, such as collapse prevention or life safety, depending on the risk category for the structure (Fig. C11.5-1). By evaluating liquefaction for the MCE rather than the design earthquake peak ground acceleration, the ground motion for the liquefaction assessment increases by a factor of 1.5. This increase in peak ground acceleration to the MCE level means that sites that previously were nonliquefiable could now be liquefiable, and sites where liquefaction occurred to a limited extent under the design earthquake could undergo more liquefaction, in terms of depth and lateral extent. Some mechanisms that are directly related to the development of liquefaction, such as lateral spreading and flow or ground settlement, could also increase in severity.

This change in peak ground acceleration level for the liquefaction evaluation addressed an issue that has existed and has periodically been discussed since the design earthquake concept was first suggested in the 1990s. The design earthquake ground motion was obtained by multiplying the MCE ground motion by a factor of 2/3 to account for a margin in capacity in most buildings. Various calibration studies at the time of code development concluded that for the design earthquake, most buildings had a reserve capacity of more than 1.5 relative to collapse. This reserve capacity allowed the spectral accelerations for the MCE to be reduced using a factor of 2/3, while still achieving safety from collapse. However, liquefaction potential is evaluated at the selected MCE_G peak ground acceleration and is typically determined to be acceptable if the factor of safety is greater than 1.0, meaning that there is no implicit safety margin on liquefaction potential. By multiplying peak ground acceleration by a factor of 2/3, liquefaction would be assessed at an effective return period or probability of exceedance different than that for the MCE. However, ASCE 7-10 requires that liquefaction be evaluated for the MCE.

Item 3 of Section 11.8.3 of the ASCE 7-10 standard lists the various potential consequences of liquefaction that must be assessed; soil downdrag and loss in lateral soil reaction for pile foundations are additional consequences that have been included in this paragraph. This section of the new provisions, as in previous editions, does not present specific seismic criteria for the design of the foundation or substructure, but item 4 does state that the geotechnical report must include discussion of possible measures to mitigate these consequences.

A liquefaction resource document has been prepared in support of these revisions to Section 11.8.3. The resource document "Evaluation of Geologic Hazards and Determination of Seismic Lateral Earth Pressures," includes a summary of methods that are currently being used to evaluate liquefaction potential and the limitations of these methods. This summary appears as Resource Paper RP 12 in the 2009 NEHRP provisions (FEMA 2009). The resource document summarizes alternatives for evaluating liquefaction potential, methods for evaluating the possible consequences of liquefaction (e.g., loss of ground support and increased lateral earth pressures) and methods of mitigating the liquefaction hazard. The resource document also identifies alternate methods of evaluating liquefaction hazards, such as

analytical and physical modeling. Reference is made to the use of nonlinear effective stress methods for modeling the buildup in pore water pressure during seismic events at liquefiable sites.

Evaluation of Dynamic Seismic Lateral Earth Pressures. The dynamic lateral earth pressure on basement and retaining walls during earthquake ground shaking is considered to be an earthquake load, E , for use in design load combinations. This dynamic earth pressure is superimposed on the preexisting static lateral earth pressure during ground shaking. The preexisting static lateral earth pressure is considered to be an H load.

C11.9 VERTICAL GROUND MOTIONS FOR SEISMIC DESIGN

C11.9.2 MCE_R Vertical Response Spectrum. Previous editions of ASCE 7 do not provide adequate guidance regarding procedures for estimating vertical ground motion levels for use in earthquake-resistant design. Historically, the amplitude of vertical ground motion has been inferred to be two-thirds ($2/3$) the amplitude of the horizontal ground motion. However, studies of horizontal and vertical ground motions over the past 25 years have shown that such a simple approach is not valid in many situations (e.g., Bozorgnia and Campbell 2004, and references therein) for the following main reasons: (1) vertical ground motion has a larger proportion of short-period (high-frequency) spectral content than horizontal ground motion, and this difference increases with decreasing soil stiffness, and (2) vertical ground motion attenuates at a higher rate than horizontal ground motion, and this difference increases with decreasing distance from the earthquake. The observed differences in the spectral content and attenuation rate of vertical and horizontal ground motion lead to the following observations regarding the vertical/horizontal (V/H) spectral ratio (Bozorgnia and Campbell 2004):

1. The V/H spectral ratio is sensitive to spectral period, distance from the earthquake, local site conditions, and earthquake magnitude and is insensitive to earthquake mechanism and sediment depth;
2. The V/H spectral ratio has a distinct peak at short periods that generally exceeds $2/3$ in the near-source region of an earthquake; and
3. The V/H spectral ratio is generally less than $2/3$ at mid-to-long periods.

Therefore, depending on the period, the distance to the fault, and the local site conditions of interest, use of the traditional $2/3$ V/H spectral ratio can result in either an under- or overestimation of the expected vertical ground motions.

The procedure for defining the MCE_R vertical response spectrum in ASCE 7 is a modified version of the procedure taken from the 2009 *NEHRP Provisions*. Unlike the procedure contained in the 2009 *NEHRP Provisions*, the procedure provided in Section 11.9 is keyed to the MCE_R spectral response acceleration parameter at short periods, S_{MS} . The procedure is based on the studies of horizontal and vertical ground motions conducted by Campbell and Bozorgnia (2003) and Bozorgnia and Campbell (2004). These procedures are also generally compatible with the general observations of Abrahamson and Silva (1997) and Silva (1997) and the proposed design procedures of Elnashai (1997). The procedure has been modified to express the vertical ground motions in terms of MCE_R ground motions instead of design ground motions.

To be consistent with the shape of the horizontal design response spectrum, the vertical design response spectrum has four regions defined by the vertical period of vibration (T_v).

Based on the study of Bozorgnia and Campbell (2004), the periods that define these regions are approximately constant with respect to the magnitude of the earthquake, the distance from the earthquake, and the local site conditions. In this respect, the shape of the vertical response spectrum is simpler than that of the horizontal response spectrum.

The equations that are used to define the design vertical response spectrum are based on three observations made by Bozorgnia and Campbell (2004):

1. The short-period part of the 5% damped vertical response spectrum is controlled by the spectral acceleration at $T_v = 0.1$ s;
2. The mid-period part of the vertical response spectrum is controlled by a spectral acceleration that decays as the inverse of the 0.75 power of the vertical period of vibration ($T_v^{-0.75}$); and
3. The short-period part of the V/H spectral ratio is a function of the local site conditions, the distance from the earthquake (for sites located within about 30 mi (60 km) of the fault), and the earthquake magnitude (for soft sites).

ASCE 7 does not include seismic design maps for the vertical spectral acceleration at $T_v = 0.1$ s and does not preserve any information on the earthquake magnitudes or the source-to-site distances that contribute to the horizontal spectral accelerations that are mapped. Therefore, the general procedure recommended by Bozorgnia and Campbell (2004) was modified to use only those horizontal spectral accelerations that are available from the seismic design maps, as follows:

1. Estimate the vertical spectral acceleration at $T_v = 0.1$ s from the ratio of this spectral acceleration to the horizontal spectral acceleration at $T = 0.2$ s for the Site Class B/C boundary (i.e., the boundary between Site Classes B and C $\bar{v}_s = 2,500$ ft/s ($\bar{v}_s = 760$ m/s), the reference site condition for the 2008 U.S. Geological Survey National Seismic Hazard Maps). For earthquakes and distances for which the vertical spectrum might be of engineering interest (magnitudes greater than 6.5 and distances less than 30 mi (60 km), this ratio is approximately 0.8 for all site conditions (Campbell and Bozorgnia 2003).
2. Estimate the horizontal spectral acceleration at $T = 0.2$ s from the Next Generation Attenuation (NGA) relationship of Campbell and Bozorgnia (2008) for magnitudes greater than 6.5 and distances ranging between 1 and 30 mi (1 and 60 km) for the Site Class B/C boundary $\bar{v}_s = 2,500$ ft/s ($\bar{v}_s = 760$ m/s). The relationship of Campbell and Bozorgnia (2008), rather than that of Campbell and Bozorgnia (2003), was used for this purpose to be consistent with the development of the 2008 U.S. Geological Survey National Seismic Hazard Maps, which use the NGA attenuation relationships to estimate horizontal ground motions in the western United States. Similar results were found for the other two NGA relationships that were used to develop the seismic hazard and design maps (Boore and Atkinson 2008; Chiou and Youngs 2008).
3. Use the dependence between the horizontal spectral acceleration at $T = 0.2$ s and source-site distance estimated in Item 2 and the relationship between the V/H spectral ratio, source-site distance, and local site conditions in Bozorgnia and Campbell (2004) to derive a relationship between the vertical spectral acceleration and the mapped MCE_R spectral response acceleration parameter at short periods, S_S .
4. Use the dependence between the vertical spectral acceleration and the mapped MCE_R spectral response acceleration

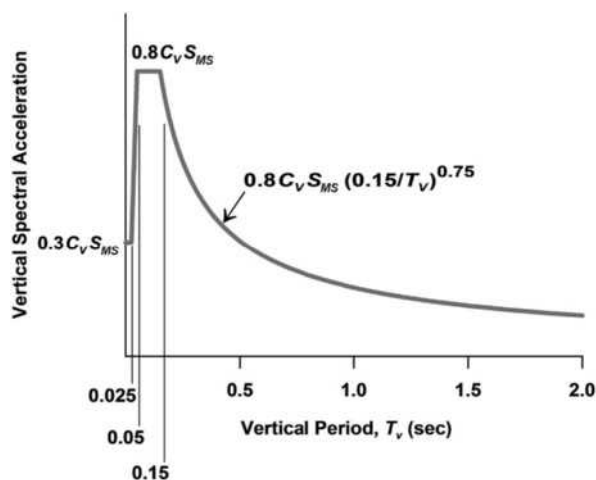


FIGURE C11.9-1 Illustrative Example of the Vertical Response Spectrum

parameter at short periods, S_S , in Item 3 to derive a vertical coefficient, C_v , that when multiplied by 0.8 and the MCE_R horizontal response acceleration at short periods, S_{MS} , results in an estimate of the design vertical spectral acceleration at $T_v = 0.1$ s.

The following description of the detailed procedure listed in Section 11.9.2 refers to the illustrated MCE_R vertical response spectrum in Fig. C11.9-1.

Vertical Periods Less Than or Equal to 0.025 s. Eq. (11.9-1) defines that part of the MCE_R vertical response spectrum that is controlled by the vertical peak ground acceleration. The 0.3 factor was approximated by dividing the 0.8 factor that represents the ratio between the vertical spectral acceleration at $T_v = 0.1$ s and the horizontal spectral acceleration at $T = 0.2$ s by 2.5, the factor that represents the ratio between the MCE_R horizontal spectral acceleration at $T = 0.2$ s, S_{MS} , and the zero-period acceleration used in the development of the MCE_R horizontal response spectrum. The vertical coefficient, C_v , in Table 11.9-1 accounts for the dependence of the vertical spectral acceleration on the amplitude of the horizontal spectral acceleration and the site dependence of the V/H spectral ratio as determined in Items 3 and 4 above. The factors are applied to S_{MS} rather than to S_S because S_{MS} already includes the effects of local site conditions.

Vertical Periods Greater Than 0.025 s and Less Than or Equal to 0.05 s. Eq. (11.9-2) defines that part of the MCE_R vertical response spectrum that represents the linear transition from the part of the spectrum that is controlled by the vertical peak ground acceleration and the part of the spectrum that is controlled by the dynamically amplified short-period spectral plateau. The factor of 20 is the factor that is required to make this transition continuous and piecewise linear between these two adjacent parts of the spectrum.

Vertical Periods Greater Than 0.05 s and Less Than or Equal to 0.15 s. Eq. (11.9-3) defines that part of the MCE_R vertical response spectrum that represents the dynamically amplified short-period spectral plateau.

Vertical Periods Greater Than 0.15 s and Less Than or Equal to 2.0 s. Eq. (11.9-4) defines that part of the MCE_R vertical response spectrum that decays with the inverse of the vertical period of vibration raised to the 0.75 power.

Two limits are imposed on the MCE_R vertical response spectrum defined by Eqs. (11.9-1) through (11.9-4) and

illustrated in Fig. C11.9-1. The first limit restricts the applicability of the vertical response spectrum to a maximum vertical period of vibration of 2 s. This limit accounts for the fact that such large vertical periods are rare (structures are inherently stiff in the vertical direction) and that the vertical spectrum might decay differently with period at longer periods. There is an allowance for developing a site-specific MCE_R vertical response spectrum if this limit is exceeded (see Section 11.4 or Chapter 21 for guidance on applying site-specific methods). The second limit restricts the MCE_R vertical response spectrum to be no less than 50% of the MCE_R horizontal response spectrum. This limit accounts for the fact that a V/H spectral ratio of one-half (1/2) is a reasonable, but somewhat conservative, lower bound over the period range of interest, based on the results of Campbell and Bozorgnia (2003) and Bozorgnia and Campbell (2004).

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